

ENCARNACION, Paraguay

WMO: 862970

Lat: 27.317S

Lon: 55.833W

Elev: 91

StdP: 100.24

Time Zone: -4.00 (PRY)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.4	5.5	0.1	3.8	7.6	1.9	4.4	9.1	10.5	23.0	8.8	21.6	1.0	50	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	35.6	24.0	34.4	23.7	33.2	23.8	26.5	32.1	25.9	31.3	25.4	30.6	4.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.2	29.8	24.3	19.5	29.1	23.7	18.8	28.5	83.6	32.6	80.6	30.9	78.5	31.4	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.9	7.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	26.8	26.0	24.9	22.0	19.1	16.8	16.7	18.0	20.0	22.7	23.4	26.0
	DBStd	5.21	2.16	2.29	2.48	3.70	4.16	4.99	5.46	5.14	4.63	3.39	2.93	2.44
	HDD10.0	17	0	0	0	0	1	6	8	2	1	0	0	0
	HDD18.3	352	0	0	0	11	42	88	100	72	33	3	2	0
	CDD10.0	4342	522	447	461	360	282	209	216	251	301	394	402	497
	CDD18.3	1634	264	213	203	121	66	42	49	62	84	138	153	238
	CDH23.3	14551	2734	1948	1736	901	383	207	288	519	742	1207	1436	2451
CDH26.7	5792	1186	794	684	306	100	35	63	181	304	491	553	1095	
Wind	WSAvg	3.1	2.8	2.7	2.7	2.8	3.0	3.2	3.2	3.3	3.3	3.6	3.3	3.1
Precipitation	PrecAvg	1794	154	153	143	176	139	133	93	102	137	198	168	175
	PrecMax	2847	327	480	422	453	582	379	352	301	379	472	557	537
	PrecMin	1099	10	8	12	0	9	7	4	6	19	24	2	10
	PrecStd	429	88	94	81	118	98	81	67	66	75	106	117	102
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	35.6	34.7	33.2	31.7	29.2	30.3	33.2	35.5	37.0	35.7	36.6
		MCWB	24.9	23.9	23.8	23.8	22.7	20.9	21.1	21.4	22.0	24.0	23.2	23.9
	2%	DB	34.9	34.0	33.2	31.9	29.3	27.6	28.4	30.9	32.9	33.8	33.6	35.2
		MCWB	24.9	24.0	24.5	23.6	21.9	20.4	20.2	20.8	21.6	23.7	22.2	23.6
	5%	DB	33.6	32.7	32.0	30.3	27.4	26.0	26.8	28.5	30.5	31.2	31.9	33.8
		MCWB	24.7	24.1	23.9	23.1	21.0	20.0	20.1	20.1	20.6	22.7	21.9	23.9
	10%	DB	32.2	31.3	30.6	28.6	25.3	24.0	24.7	26.2	27.6	29.1	30.0	32.0
		MCWB	24.4	23.8	23.4	22.3	20.8	19.0	19.1	19.3	19.4	21.8	21.2	23.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	26.9	26.7	25.5	23.7	22.9	23.4	23.4	23.6	26.0	25.5	27.0
		MCDB	33.1	31.6	32.3	30.8	28.7	26.8	28.3	29.8	32.1	33.2	30.6	32.8
	2%	WB	26.5	25.9	25.7	24.5	22.8	21.4	21.7	21.9	22.6	24.6	24.3	26.1
		MCDB	32.1	30.8	31.2	29.6	27.5	25.3	26.3	28.4	30.4	31.3	30.5	32.0
	5%	WB	26.0	25.3	25.0	23.7	22.0	20.5	20.5	20.8	21.3	23.6	23.4	25.2
		MCDB	31.3	29.9	30.0	28.5	25.9	24.5	25.0	26.8	28.1	29.8	29.3	31.0
	10%	WB	25.3	24.7	24.2	23.0	21.1	19.6	19.5	19.6	20.4	22.7	22.6	24.5
		MCDB	30.4	29.2	28.8	27.5	24.4	23.3	23.7	25.1	26.0	27.7	27.5	30.0
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.0	10.2	10.1	9.0	9.7	9.9	11.5	10.8	10.3	11.3	10.7
		MCDBR	12.0	12.1	12.2	11.5	11.1	10.8	11.5	13.7	13.7	13.6	14.4	13.1
		MCWBR	5.0	5.1	5.3	5.2	5.2	5.4	5.8	6.3	5.9	6.4	6.2	5.3
	5% WB	MCDBR	9.9	9.7	10.2	10.5	9.1	9.5	10.8	12.3	12.2	11.6	10.6	10.6
		MCWBR	5.0	4.7	5.3	5.6	5.0	5.8	6.3	6.8	6.0	6.1	5.7	5.3
Clear-Sky Solar Irradiance	taub	0.414	0.410	0.400	0.394	0.373	0.395	0.394	0.493	0.560	0.470	0.411	0.416	
	taud	2.392	2.403	2.418	2.406	2.430	2.367	2.345	2.018	1.860	2.173	2.351	2.366	
	Ebn at Noon	930	917	895	850	822	772	789	738	737	852	926	931	
	Edh at Noon	128	124	116	107	95	97	102	155	199	154	133	132	
All-Sky Solar Radiation	RadAvg	6.58	6.09	5.29	4.23	3.14	2.65	3.04	3.67	4.36	5.22	6.37	6.68	
	RadStd	0.50	0.45	0.36	0.41	0.42	0.23	0.33	0.28	0.47	0.70	0.63	0.61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+74

Nomenclature: See separate page